EXPERIMENT 2

```
#include <stdio.h>
// FCFS Scheduling
void fcfs(int n, int bt[]) {
int wt[10], tat[10];
wt[0] = 0;
for(int i = 1; i < n; i++)
wt[i] = wt[i-1] + bt[i-1];
printf("\nFCFS Scheduling:\n");
printf("Process\tBurst Time\tWaiting Time\tTurnaround Time\n");
for(int i = 0; i < n; i++) {
tat[i] = wt[i] + bt[i];
printf("P%d\t\t%d\t\t%d\t\t%d\n", i+1, bt[i], wt[i], tat[i]);
}
}
// Round Robin Scheduling
void roundRobin(int n, int bt[], int tq) {
int rem_bt[10], wt[10], tat[10];
int t = 0;
for(int i = 0; i < n; i++) {
rem_bt[i] = bt[i];
wt[i] = 0;
}
while(1) {
int done = 1;
for(int i = 0; i < n; i++) {
if(rem_bt[i] > 0) {
```

```
done = 0;
if(rem_bt[i] > tq) {
t += tq;
rem_bt[i] -= tq;
} else {
t += rem_bt[i];
wt[i] = t - bt[i];
rem_bt[i] = 0;
}
}
}
if(done == 1)
break;
}
printf("\nRound Robin Scheduling (Time Quantum = %d):\n", tq);
printf("Process\tBurst Time\tWaiting Time\tTurnaround Time\n");
for(int i = 0; i < n; i++) {
tat[i] = bt[i] + wt[i];
printf("P%d\t\t%d\t\t%d\t\t%d\n", i+1, bt[i], wt[i], tat[i]);
}
}
int main() {
int n, tq;
int bt[10];
printf("Enter number of processes: ");
scanf("%d", &n);
printf("Enter burst times:\n");
for(int i = 0; i < n; i++) {
printf("P%d: ", i+1);
```

```
scanf("%d", &bt[i]);
}
fcfs(n, bt);
printf("\nEnter Time Quantum for Round Robin: ");
scanf("%d", &tq);
roundRobin(n, bt, tq);
return 0;
}
```

cocalc.com/projects/791d6841-a63b-4f4b-92f2-4e08fd664bad/files/2026-02-10-file-1.term

Projects | My First Project

Hello Buy a license or redeem a voucher.
Prices start at a few $ per month. Apply your license to this project. Free projects...

2026-02-10-file-1.term | Priority.c

Assistant | Server | ~ | Terminal | File | Edit | View | Go | Help

Explorer | New | Log | Find | Servers | Users | Upgrades | Processes | Settings

```
~$ ./Priority
Enter number of processes: 4
Enter Burst Time and Priority for P1: 4
6
Enter Burst Time and Priority for P2: 6

6
Enter Burst Time and Priority for P3: 8
8
Enter Burst Time and Priority for P4: 9
9

Process BT      Priority        WT      TAT
P1      4       6               0       4
P2      6       6               4       10
P3      8       8               10      18
P4      9       9               18      27
~$ gcc Priority.c -o Priority
~$ ./Priority
Enter number of processes: 4
Enter Burst Time and Priority for P1: 2 1
Enter Burst Time and Priority for P2: 3 2
Enter Burst Time and Priority for P3: 4 3
Enter Burst Time and Priority for P4: 4 4

Process BT      Priority        WT      TAT
P1      2       1               0       2
P2      3       2               2       5
P3      4       3               5       9
P4      4       4               9       13
~$
```